

Package #160

apply a marine■inspired anti■biofouling micro■textured surface (e.g., shark■skin riblet pattern) to the catheter lumen in conjunction with the existing antithrombotic coating, creating a physical barrier that hinders protein and platelet adhesion that leads to heparin■binding thrombus

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'heparin-binding thrombus formation' concerning tunneled hemodialysis catheter. User-submitted engineering problem (session input): "Why do tunneled hemodialysis catheters develop heparin-binding thrombus despite antithrombotic luminal coatings?". [fda_maude] 100 records retrieved live (1992-2070). Illustrative records: 'MAUDE report 1000: Injury [SAME AS ABOVE]'; 'MAUDE report 1000009: Injury [SLIP-CATH BEACON TIP CATHETER]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2025-2026). Illustrative records: 'Antithrombotic Polymers: A Narrative Review on Current and Future Strategies for Their Design, Synthesis, and Application.'; 'Polymeric Fibrous Materials for Procoagulant and Anticoagulant Applications: A Review of Molecular Blood-Material Mechanisms and Strategies.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: biofilm formation

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
- 03 - Buyer Decision Card
- 04 - Evidence Summary
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Status: ENGINEERING_DEFINITION. Not physically validated.